FILTER DESIGN TEST CASE

Multiplexer Module - Band 5 (850 MHz) for Millimeter-Wave Beamforming
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Multiplexer Module filter targeting Band 5 (850 MHz) for Millimeter-Wave Beamforming. The filter is designed on 42° YX $LiTaO_3$ substrate with a target center frequency of 2641.7 MHz and bandwidth of 33.9 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Millimeter-Wave Beamforming module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Multiplexer Module
- Frequency Band: Band 5 (850 MHz)
- Application: Millimeter-Wave Beamforming
- Substrate: 42° YX $LiTaO_3$
- Package: WLP (Wafer Level Package)
- Design Tool: PathWave EM Design
- Design Center: Helsinki 5G Lab

This specification applies to all variants of the Multiplexer Module design targeting Band 5 (850 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 2641.7 MHz
Bandwidth (3 dB): 33.9 MHz
Insertion Loss Target: <= 3.2 dB
Return Loss Target: >= 20.0 dB
Out-of-Band Rejection: >= 34.0 dB
Isolation (Duplexer): >= 27.0 dB
Q Factor: 1591
Temperature Coefficient: -27.6 ppm/°C
Die Size: 1.8 x 2.2 mm
Wafer Size: 6-inch
Number of Resonators: 7
Electrode Material: Mo

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open PathWave EM Design and load the Multiplexer Module template project.
2. Define the substrate stack: 42° YX LiTaO3 with specified layer thicknesses.
3. Set design targets: center freq 2641.7 MHz, BW 33.9 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (WLP (Wafer Level Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, 42° YX LiTaO3).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 7925 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.2 dB
- Return loss in passband shall be >= 20.0 dB
- Out-of-band rejection at +/- 51 MHz offset >= 34.0 dB
- Group delay variation across passband shall be <= 2 ns
- Temperature drift over -40°C to +85°C shall be <= 9.1 MHz
- Power handling shall be >= +28 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 95%

6. TEST RESULTS

Measurement Date: 2025-04-05
Devices Tested: 30
Insertion Loss (meas): 2.86 dB
Return Loss (meas): 21.9 dB
Rejection (meas): 36.5 dB
Isolation (meas): 32.3 dB
Q Factor (meas): 1591
Group Delay Variation: 2.4 ns
Power Handling: +26 dBm
Die Yield: 95.3%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

a) The Multiplexer Module design demonstrated acceptable performance for Band 5 (850 MHz).
b) Insertion loss is marginally above target; electrode thickness optimization recommended.
c) Out-of-band rejection exceeded the minimum requirement.
d) Package parasitic effects were consistent with 3D EM model predictions.
e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

1. Review near-band rejection margin for worst-case temperature.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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